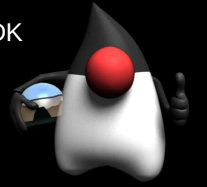


Java Programming I
Lecture 2C
The Java 6 Development Kit

Installing the JDK
on Windows



Slide 1

Java Programming I

Lecture 2C

The Java 6 Development Kit

Duration: 00:00:19

Advance mode: Auto

Notes:

This presentation will show you how to download and install the JDK 6 software. These instructions are for Windows XP or Windows Vista. If you're using Linux the process is very similar. If you have a Macintosh computer, the JDK should already be installed and ready to use.

The Java Development Kit

- Software used to write Java programs
 - Not needed to run Java programs
- Different versions of Java
 - Latest version is Java 6, update 6
 - Previous version is Java 5
- External and internal names
 - JDK 1.5 is Java 5, JDK 1.6 is Java 6

Slide 2

The Java Development Kit

Duration: 00:00:59

Advance mode: Auto

Notes:

The JDK, or Java Development Kit is software that programmers use when writing Java programs. Even if your computer can already run Java programs, you'll need to install the JDK if you want to create your own programs.

The JDK comes in several different versions. The latest one available when I wrote this was Java 6, update 3. A later update may be available when you see this. (On the Mac, just run Software Update to automatically install the latest version if necessary.)

There are actually several different versions of Java, and the versions are confusing because there is both an internal and an external number. Names like Java 5 and Java 6 are the external names; internally, the versions will be known as JDK 1.5 or JDK 1.6. Both of those mean the same thing. Let's go ahead and install Java 6, or JDK 1.6.

The Sun Java Download Page

- <http://java.sun.com/javase/downloads/>



Slide 3

Notes:

Start your download by visiting Sun Microsystems download page at <http://java.sun.com/javase/downloads/>. This will always take you to a page where you can download the latest version of the base or standard edition of Java. (That's what the se means).

When you get to the downloads page, locate the appropriate section for downloading the JDK or Java Development Kit. In my example here, it's the first download, but that may not always be the case. (For instance, this summer the first entry is for a beta version of the JDK, while the latest released version of the JDK, update 6, is the second entry.) You don't want any other bundles, such as NetBeans, Java EE and you

The Sun Java Download Page

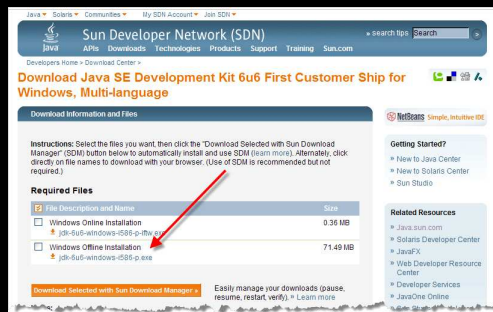
Duration: 00:01:32

Advance mode: Auto

especially want to make sure you don't download the JRE or Java Runtime Environment. The JRE allows you to run Java programs, but does include all of the tools that you'll need to write your own. When you download and install the JDK, it will install the latest JRE as well, so you won't need to do that separately.

Once you've located the correct link, click the download button on the right, and you'll be taken to a page with downloads for the various platforms. (If you use Internet Explorer, you'll also see a couple of irritating messages about secure and insecure data, which you'll need to click through.)

License and Download



Slide 4

License and Download

Duration: 00:01:08

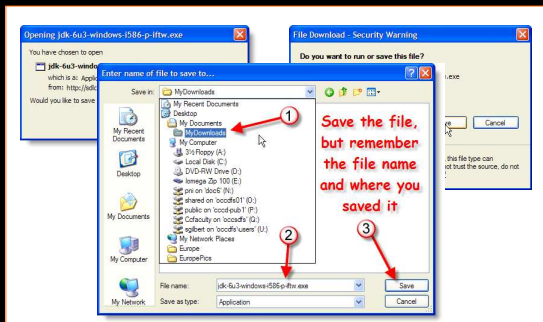
Advance mode: Auto

Notes:

On the download page, the first thing you need to do is to select the version of the JDK you want to download, (Step 1 in the picture here), and then accept the license agreement by clicking the little checkbox shown here for Step 2; you can't actually download anything until you do, so you might as well go ahead. I've never actually read the agreement myself, but I'm sure that it's not that onerous. (On the other hand, maybe I've just signed away my bank account!)

Once you've agreed to the license, you can click the Continue to begin your download. On the next screen you'll be allowed to click the hyperlink to download the installation program. There are two versions of the Windows SDK download. One version downloads everything to your local machine and installs it from there, while one version downloads a small installer program, that downloads only the files needed while the installation is running. I've used both of them without problem. Some students, though, have reported difficulty with the smaller "online" installation, so my instructions here will show the "offline"

Save the File



Slide 5

Save the File

Duration: 00:00:44

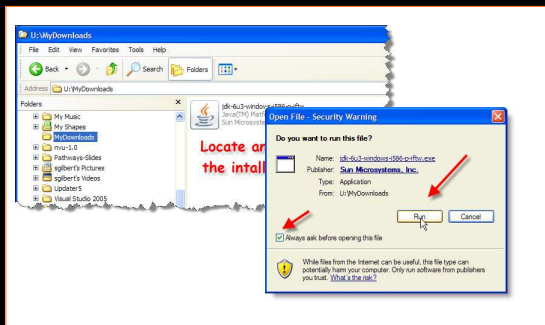
Advance mode: Auto

Notes:

If you're using Firefox (which I recommend), you'll see a dialog that asks you to save the file. If you're using Internet Explorer you'll see a similar dialog, along with some dire warnings about damaging your machine. You'll also see a button that allows you to Run the file; don't do that. Whichever dialog you see, make sure that you just save the file on your local machine.

When the Save dialog appears (assuming that you haven't left your browser automatically downloading to the desktop), make sure that you remember both where you've saved the file and what the file was named. You'll need that information after the download has completed, which, depending on your Internet connection, could be awhile.

Run the Installer



Slide 6

Run the Installer

Duration: 00:00:52

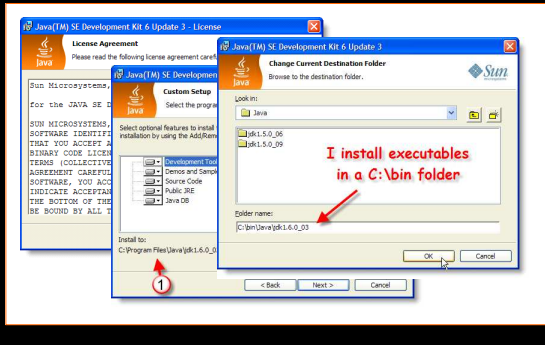
Advance mode: Auto

Notes:

To install the software, locate the file you downloaded in Windows Explorer (Win-E is the quick shortcut) and double-click it to start the installation. If you downloaded the file using Internet Explorer, you'll also get one more Security Warning dialog to click through, before you can run the file.

In Windows Vista, you'll probably be asked to supply your administrator password; in Windows XP, if you aren't logged in as an administrator, you might be asked if you want to run the program as an administrator. You can't install the JDK as a "regular user" without the administrator password. (However, if you want to do that, just bring a USB thumb drive and copy the JDK folder from one of the machines in the OCC lab and then copy it to your machine, which you can do as a regular user.)

License and Directories



Slide 7

License and Directories

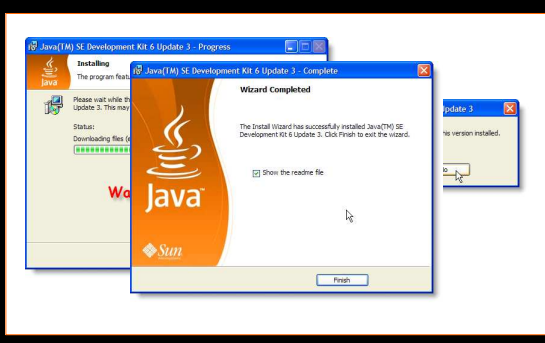
Duration: 00:00:44

Advance mode: Auto

Notes:

When the installer program starts up, it first asks you to agree to another license (which you can't skip, naturally), and then takes you to the Custom Setup screen. I'd just leave all of these options as they are, unless you want to install the program in a different directory. To do that, just click the Change button and navigate to the directory where you want the program installed. I, for instance, install all of my programs in a folder that I've created named C:\bin\Java. I'd suggest that when you first start out, though, you just use the default location which currently uses C:\Program Files\Java along with separate subdirectories for each version of the JDK.

Installation and the JRE



Slide 8

Installation and the JRE

Duration: 00:00:43

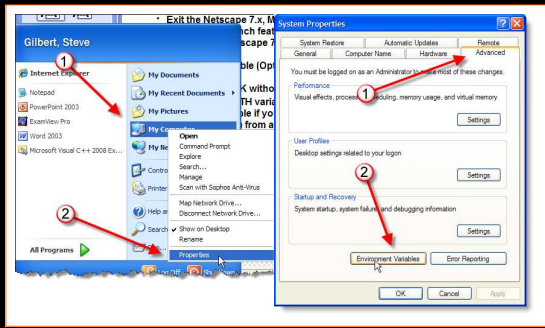
Advance mode: Auto

Notes:

Now, you need to wait a bit. The dialog says that it may take several minutes to download files, but the Offline installer doesn't actually download any files from the Internet; I think that they just used the same messages for both the online and offline installer programs. On my computers, this part usually takes between five and ten minutes, which isn't bad.

After the JDK is installed, a second installer program will run to install the JRE or Java Runtime Environment. If you already have the latest JRE (perhaps from visiting <http://www.java.com>) then you'll see a little message telling you so, and you don't need to install the JRE a second time.

Set the PATH



Slide 9

Set the PATH

Duration: 00:01:10

Advance mode: Auto

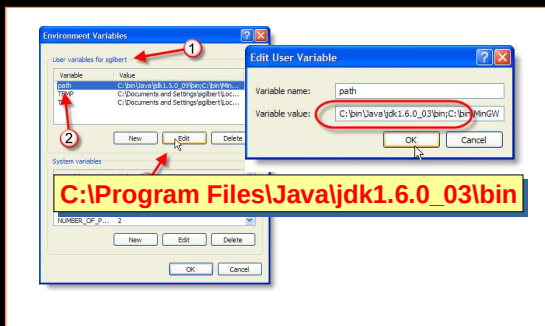
Notes:

Once the installation is complete, there is one more step you'll need to do manually to make the JDK a little easier to use: you'll need to add the JDK bin directory to your computer's executable path. The executable path is the set of directories that your computer searches for files when you tell it to run a program using the Run command or the Command shell window. Of course, this part isn't absolutely necessary, especially if you always use integrated Java development tools like BlueJ or Eclipse. It does make it easier to write and run Java programs using just a simple text editor though.

To set the Path, locate My Computer on the Start menu, or on the desktop, right-click the icon, and choose Properties from the context menu. You can also double click the System icon in the Control Panel.

On the System Properties dialog first locate the Advanced tab and then click the Environmental Variables tab near the bottom of the screen.

The PATH User Variable



Slide 10

The PATH User Variable

Duration: 00:01:29

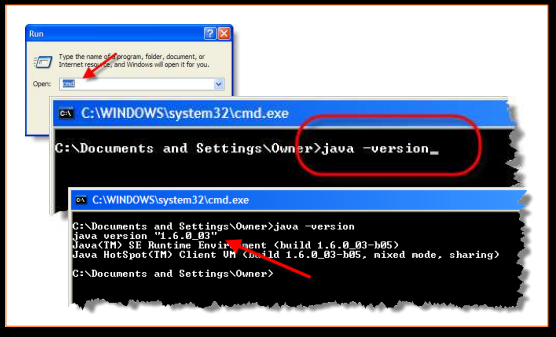
Advance mode: Auto

Notes:

If there is already a variable named PATH, then click the Edit button. If there is no variable named PATH, click the New button and create one. (I don't recommend editing the PATH variable in the System variables section, since it seems pretty easy to accidentally delete an entry that is important for your computer's operation).

When the little Edit User Variable window appears, replace any existing JDK entry with the location of the bin folder in the JDK you just installed. In my case it was C:\bin\Java\jdk1.6.0_03\bin. Of course, by the time you see this, the JDK version number (the _03) part will be different, and, if you use the default installation, the path will probably be C:\Program Files\Java\jdk1.6.0_??\bin, with the two question marks being the update revision number. If the path setting already contains other entries, don't erase them. Just type the new path before the existing entries and follow it with a semicolon, leaving the existing path items as they were, following your new entry.

Testing your Installation



Slide 11

Testing your Installation

Duration: 00:01:05

Advance mode: Auto

Notes:

To make sure you've set the path correctly, open the Run dialog, type CMD and press Enter. In the Command Window, type `java -version`. Put a space between the word `java` and `-version`, but don't put a space between the dash or hyphen and the word `version`. (This is called a "switch" and it tells the program which action to perform.) Press the Enter key and you should see a response telling you that you have the correct version installed.

Now, do the same thing, but type the word `javac` (rather than `java`). This is the Java compiler. You should get another, different message printing the compiler version as well. If you get a message about an unknown option, file not found, or something like that, then go back over the steps for setting the PATH. If you still have problems, then contact me, and we'll work it out.

That's it! You're all done installing the Java Development Kit.